

University of Gujrat

Department of Software Engineering

Cost-Benefit Analysis and Risk Evaluation

Submitted to:

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Subject:

Information Security

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Scenario Overview

CampusMart is a small online store that keeps all customer orders and payment records on a single database server. The IT manager is worried about ransomware.

Calculations & Solutions

1. Calculate the SLE and ALE before the safeguard.

Asset Value (AV): 2,000,000 PKR
Exposure Factor (EF): 60% (0.60)
Single Loss Expectancy (SLE): AV × EF = 2,000,000 × 0.60 = 1,200,000 PKR
Annual Rate of Occurrence (ARO): 1 attack / 2 years = 0.5
Annual Loss Expectancy (ALE): SLE × ARO = 1,200,000 × 0.5 = 600,000 PKR

2. Calculate the Annual Cost of the Safeguard (ACS).

Annual Purchase Cost (APC): 150,000 / 3 years = 50,000 PKR
Annual License Fee (ALF): 60,000 PKR
Annual Admin and Maintainence Labor (AAML): 30,000 PKR
Annual Cost of Safeguard (ACS): APC + ALF + AAML = 50,000 + 60,000 + 30,000 = 140,000 PKR

3. Calculate the SLE and ALE after the safeguard.

Exposure Factor After (EF_{after}): 10% (0.10)

Annual Rate of Occurrence After (ARO_{after}): 1 in 5 years = 0.20

Single Loss Expectancy After (SLE_{after}): AV × EF_{after} = 2,000,000 × 0.10 = 200,000 PKR

Annual Loss Expectancy After (ALE_{after}): SLE_{after} × ARO_{after} = 200,000 × 0.20 = 40,000 PKR

4. Calculate the annual risk reduction and the net benefit of the safeguard.

Annual Risk Reduction (ARR): ALE - ALE_{after} = 600,000 - 40,000 = 560,000 PKR

Net Benefit (NB): ARR - ACS = 560,000 - 140,000 = 420,000 PKR

5. Calculate the Cost-Benefit Ratio and ROI.

Cost-Benefit Ratio (CBR): ACS / ARR = 140,000 / 560,000 = 0.25
Return on Investment (ROI): (NB / ACS) × 100 = (420,000 / 140,000) × 100 = 300%

6. Calculate the payback period.

Initial Investment: 150,000 PKR
Annual Operating Expenses: 60,000 (License) + 30,000 (Admin) = 90,000 PKR
Annual Net Cash Flow (ANCF): ARR - Annual Operating Expenses = 560,000 - 90,000 = 470,000 PKR
Payback Period: Initial Investment / ANCF = 150,000 / 470,000 = 0.319 years (~3.8 months)

7. Should CampusMart buy the safeguard? Justify your answer.

Recommendation: Yes, CampusMart should definitely purchase the safeguard.

Justification: The safeguard gives a great Net Benefit of 420,000 PKR per year and an excellent Return on Investment (ROI) of 300%. The Cost-Benefit Ratio of 0.25 shows that for every 1 PKR spent, 4 PKR is saved. Also, the initial investment will be fully recovered in just ~3.8 months (0.319 years). This makes it a financially great and highly effective decision to purchase this safeguard.